

User Experience Design - Complete Exam Question Bank

MODULE 1: Introduction to User Interface Design

A. Difference Between Questions

Q1.1 Compare User Interface (UI) and User Experience (UX) Design.

Answer:

UI Design	UX Design
UI is what a user sees	UX is how a user feels
Focuses on visual elements and layout	Focuses on overall user interaction and satisfaction
Responsible for designing screens users interact with	Makes sure the interaction is pleasurable
Chooses components and interaction styles	Works on factors like latency, overall time to perform operations
Deals with appearance and functionality	Deals with effectiveness and user emotions
A good UI contributes to better UX	UX encompasses UI and much more

Q1.2 Differentiate between External Consistency and Internal Consistency in design.

Answer:

- **External Consistency:** Refers to the appearance of screens and components being consistent with industry standards and other similar applications
- **Internal Consistency:** Refers to storage, data structure, and format being consistent throughout the same application/system

B. Short Answer Questions

Q1.3 Define User Interface and list its main interaction styles.

Answer: User Interface is the layout or structure of elements on the screen as well as the functional behavior of such elements when manipulated by the user. The UI design should be user-friendly.

Main interaction styles:

1. Command line
2. Graphical User Interface (GUI)
3. Voice Activated
4. Gesture based
5. Brain Controlled
6. Others (face recognition, biometric, iris)

Q1.4 What is User Experience? List the expectations users have from a system.

Answer: User Experience refers to how a user feels while/after using the system. It's not restricted only to computer systems but all devices/gadgets used these days.

User expectations:

- Intuitive interface
- Easy to learn
- Allow to make errors (with recovery)
- Look beautiful
- Responsive and efficient

Q1.5 List the 5 stages of the UX Design Process.

Answer:

1. Researching - Gather fundamental information about target users and their needs
2. Analyze - Plan and map solutions, visualize feasible content
3. Designing - Develop the product and create multiple versions
4. Testing - Test with real users to identify issues
5. Launching - Release to public and gather feedback

C. Long Answer Questions

Q1.6 Explain the User-Centered Design (UCD) process with its phases.

Answer: User-centered design (UCD) is an iterative design process in which designers focus on users and their needs in each phase of the design process. It involves users throughout the design process via various research and design techniques to create highly usable and accessible products.

Phases in User-Centered Design:

1. **Understand context of use:** Mental state of user, psychology of the user, application of use
2. **Specify user requirements:** Research to find what difficulties/lacunas users are facing in existing systems
3. **Design solution:** Implement ideas in multiple styles/ways to resolve issues
4. **Evaluate against requirements:** Usability and User Experience testing of the solutions

This is an iterative process where feedback from each phase informs improvements in subsequent iterations.

Q1.7 Describe the key design principles for UX design with examples.

Answer: Key UX Design Principles:

1. **Focus on the user:** Study background of end users including gender, age, physical parameters, educational background, cultural background, and previous experience
2. **Consistency:** Maintain both external consistency (appearance matching industry standards) and internal consistency (uniform data structure and storage)
3. **Context is key:** Understand the background context and user's previous knowledge
4. **Put the user in control:** System should be designed to let users control their actions rather than the system controlling users
5. **Accessibility:** Easy to use for all types and ages of users, multilingual support, effective usage for people with special needs, compatibility with other systems
6. **Usability testing:** Continuous testing with real users to validate design decisions

D. Application-Based Questions

Q1.8 You are designing a mobile banking application. Apply the principle of "Put the user in control" to design the transaction flow. Explain your design decisions.

Answer: Application of "Put the user in control" for mobile banking:

1. **Transaction Confirmation:** Before any transaction, show clear summary with amount, recipient, and purpose. Allow users to review and confirm/cancel
2. **Multiple Authentication Options:** Provide choices like PIN, fingerprint, face recognition, or OTP based on user preference
3. **Transaction Limits:** Allow users to set their own daily/monthly transaction limits
4. **Undo/Cancel Options:** Provide clear cancel buttons at each step before final submission
5. **Progress Indicators:** Show users exactly where they are in the transaction process
6. **Backup Options:** If one method fails (e.g., fingerprint), immediately offer alternatives
7. **Clear Exit Points:** Always provide visible "Cancel" or "Back" options

Design Rationale: Users feel more confident when they control the pace and can exit anytime. This reduces anxiety and increases trust in financial applications.

Q1.9 Analyze a popular e-commerce website's checkout process using the accessibility principle. Suggest improvements.

Answer: Accessibility Analysis of E-commerce Checkout:

Current Issues:

- Small text sizes (below 16px) difficult for elderly users
- Color-only indicators for errors (problematic for color-blind users)
- Complex forms requiring too much information
- No keyboard navigation support
- Missing alt text for product images

Improvements:

- 1. **Font Size:** Minimum 16px for body text, 24px+ for headings
 - 2. **Color Contrast:** Ensure WCAG 2.1 AA compliance with 4.5:1 contrast ratio
 - 3. **Multiple Error Indicators:** Use color, icons, and text together
 - 4. **Keyboard Navigation:** All interactive elements accessible via Tab key
 - 5. **Screen Reader Support:** Proper ARIA labels and semantic HTML
 - 6. **Multilingual Support:** Option to switch languages
 - 7. **Simplified Forms:** Progressive disclosure, asking only essential information initially
 - 8. **Clear Instructions:** Simple language avoiding jargon
-

MODULE 2: User Research and Journey

A. Difference Between Questions

Q2.1 Compare the Five Planes of User Experience (Strategy vs Surface).

Answer:

Strategy Plane	Surface Plane
Most abstract level	Most concrete level
Defines what users and organization want	Defines what users actually see
Includes product objectives and user needs	Includes visual design, images, text
Foundation for all other planes	Final manifestation of all planes
Long-term planning	Immediate visual impact
Involves business goals and user research	Involves graphic design and interface elements

Q2.2 Differentiate between Qualitative and Quantitative User Research.

Answer:

Qualitative Research	Quantitative Research
Focused on descriptions and insights	Focused on numbers and mathematical calculations
Direct assessment based on observation	Statistical analysis of user behavior
Understanding beliefs and practices	Measuring specific metrics
Methods: interviews, ethnographic studies	Methods: analytics, A/B testing, surveys
Answers "why" and "how"	Answers "what" and "how many"
Small sample size, deep insights	Large sample size, statistical significance
Emotions and motivations	Time, clicks, conversion rates

B. Short Answer Questions

Q2.3 Define the Four Types of Users.

Answer:

1. **Achievers:** Motivated by achieving goals, enjoy overcoming obstacles and completing tasks
2. **Socializers:** Motivated by social interactions, enjoy collaborating and forming communities
3. **Players:** Focused on competition and winning
4. **Explorers:** Motivated by discovery and understanding how things work

Q2.4 List the key components of a Problem Statement for UX design.

Answer: A problem statement should include:

- Target user identification
- User's goals and objectives
- Current pain points or difficulties
- Context of use
- Desired outcomes
- Constraints and limitations

Q2.5 What are User Personas and why are they important?

Answer: User Personas are partly fictional characters created to represent different kinds of customers or target audience that will use a product or service. They are based on real data and user interviews, though personal details may be fictional.

Importance:

- Help teams understand user needs
- Guide design decisions
- Create empathy for users
- Ensure consistent user focus across teams
- Validate design choices against user goals

C. Long Answer Questions

Q2.6 Explain the Five Planes Model in detail with examples.

Answer: The Five Planes Model by Jesse James Garrett:

1. Strategy Plane (Foundation):

- **Purpose:** Define what organization and users want from the product
- **Components:** Product objectives, user needs, business goals
- **Example:** For a food delivery app - Business wants profit, users want quick food delivery

2. Scope Plane:

- **Purpose:** Define features and functions that will be included
- **Components:** Functional specifications, content requirements
- **Example:** Features like restaurant search, ordering, payment, tracking

3. Structure Plane:

- **Purpose:** Define how users navigate and how information is organized
- **Components:** Information architecture, interaction design
- **Example:** How users move from restaurant selection to payment completion

4. Skeleton Plane:

- **Purpose:** Define placement of buttons, controls, and content blocks
- **Components:** Interface design, navigation design, information design
- **Example:** Where to place "Add to Cart" button, search bar location

5. Surface Plane:

- **Purpose:** Final visual design that users see
- **Components:** Visual design, typography, colors, images
- **Example:** App's color scheme, logo placement, button styles

Development Approach: Bottom-up approach where each plane influences the one above it. Changes in upper planes may require rethinking lower planes (ripple effect).

Q2.7 Describe the comprehensive User Research process including goals, methods, and outcomes.

Answer: User Research Process:

Goals:

- Put design project into context
- Help understand the problem to be solved
- Identify who users are and their context
- Determine what users need from the designer

Research Methods:

1. User Interviews:

- One-on-one discussions with users
- Detailed information about attitudes and experiences
- Observation in natural environment

2. Surveys:

- Series of questions to multiple users
- Learn about user demographics and preferences
- Scalable data collection

3. Focus Groups:

- Moderated discussion with group of users
- Insight into user attitudes and desires
- Group dynamics and consensus

4. A/B Testing:

- Compare two different designs
- Measure user responses statistically
- Data-driven design decisions

5. Heuristic Analysis:

- Expert evaluation against usability principles
- Quick identification of usability issues
- Cost-effective problem discovery

Outcomes:

- Clear understanding of user needs
- Validated design assumptions
- Data-driven design decisions
- Reduced development costs
- Better user satisfaction

D. Application-Based Questions

Q2.8 You are designing a mobile app for farmers to check weather and crop prices. Create user personas for this application (at least 2 different personas).

Answer: Persona 1: Traditional Farmer - Ramesh

- **Age:** 45
- **Education:** Primary school
- **Tech Experience:** Basic mobile phone usage
- **Primary Crops:** Rice, wheat
- **Farm Size:** 5 acres
- **Goals:** Check daily weather, compare crop prices in nearby markets
- **Pain Points:** Difficulty reading small text, unfamiliar with complex interfaces
- **Usage Context:** Early morning and evening, often in fields with poor connectivity
- **Preferred Language:** Regional language (Hindi/local)

Persona 2: Progressive Young Farmer - Priya

- **Age:** 28
- **Education:** Agricultural degree
- **Tech Experience:** Smartphone user, social media active
- **Primary Crops:** Organic vegetables, cash crops
- **Farm Size:** 10 acres
- **Goals:** Detailed weather forecasts, market trends, new farming techniques
- **Pain Points:** Wants accurate long-term predictions, needs offline access
- **Usage Context:** Throughout day, uses for planning and decision making
- **Preferred Language:** English + regional language

Design Implications:

- Simple, icon-heavy interface for Ramesh
- Voice-based inputs and outputs
- Offline functionality for both
- Multiple language support
- Large, clear typography

Q2.9 Conduct a heuristic evaluation of a popular social media platform's mobile app. Identify violations and suggest improvements.

Answer: Heuristic Evaluation of Social Media Mobile App:

1. Visibility of System Status:

- **Violation:** No clear indication when posts are uploading
- **Improvement:** Show progress bars for uploads, clear status indicators

2. Match between System and Real World:

- **Good:** Heart icon for likes, familiar metaphors
- **Improvement:** Use more culturally relevant icons for different regions

3. User Control and Freedom:

- **Violation:** Difficult to undo accidental story posts
- **Improvement:** Add "Undo" option for recent actions, clearer exit options

4. Consistency and Standards:

- **Violation:** Different swipe gestures across features
- **Improvement:** Standardize swipe interactions throughout app

5. Error Prevention:

- **Violation:** Easy to accidentally share posts to wrong audience
- **Improvement:** Add confirmation dialog for public posts, default to restricted sharing

6. Recognition Rather than Recall:

- **Good:** Visual timeline shows post history
- **Improvement:** Add search suggestions based on previous searches

7. Flexibility and Efficiency:

- **Good:** Shortcuts for power users
- **Improvement:** Customizable quick action buttons

8. Aesthetic and Minimalist Design:

- **Violation:** Cluttered interface with too many features visible
- **Improvement:** Progressive disclosure, cleaner layout

9. Error Recovery:

- **Good:** Clear error messages for connectivity issues
- **Improvement:** Better guidance for account recovery

10. Help and Documentation:

- **Violation:** Help section is hard to find
- **Improvement:** Contextual help, onboarding tutorials

MODULE 3: UX Design Process

A. Difference Between Questions

Q3.1 Compare Individual Empathy Maps vs Aggregated Empathy Maps.

Answer:

Individual Empathy Maps	Aggregated Empathy Maps
Based on single user interview	Combines multiple individual empathy maps
Represents one specific user	Represents a user segment
Detailed personal insights	Synthesized themes across user group
Used for deep user understanding	Used for persona creation
Direct quotes from one person	Common patterns across users
Specific behavioral observations	Generalized behavioral trends

Q3.2 Differentiate between Wireframes and Prototypes in UX design process.

Answer:

Wireframes	Prototypes
Visual representation of skeletal framework	More lifelike model of the product
Static, low-fidelity	Interactive, higher fidelity
Tests information architecture	Tests interaction and functionality
Shows structure and layout	Shows behavior and user flow
Early stage design tool	Later stage validation tool
Focuses on content placement	Focuses on user interaction

B. Short Answer Questions

Q3.3 Define the 4 quadrants of an Empathy Map.

Answer:

1. **Says:** Direct quotes from user research and interviews
2. **Thinks:** What user is thinking but might not say out loud
3. **Feels:** User's emotional state during product experience
4. **Does:** Physical actions and behaviors of the user

Q3.4 What are Wicked Problems in UX Design? List their characteristics.

Answer: Wicked Problems are social or cultural problems that are difficult or impossible to solve due to their complex and interconnected nature.

Characteristics:

1. Problem is never solved completely
2. Every problem is unique
3. There is no unique solution
4. Multi-causal, multi-scalar, and interconnected
5. Multiple stakeholders with conflicting agendas
6. Allowing multidisciplinary activities
7. Every wicked problem is connected to others
8. Every solution spreads throughout systems
9. Solutions are not right/wrong but better/worse
10. Can take long time to evaluate solutions

Q3.5 List the 5 stages of Design Thinking process.

Answer:

1. **Empathize:** Understand user needs through research and observation
2. **Define:** Identify needs and problems clearly
3. **Ideate:** Generate ideas through brainstorming
4. **Prototype:** Transform ideas into tangible solutions
5. **Test:** Test the solution with users

C. Long Answer Questions

Q3.6 Explain the 6 Stages of UX Design Process in detail.

Answer: 6 Stages of UX Design Process:

1. Defining and Understanding Your Product:

- Main focus: Define how final product should work
- Interviews with key stakeholders across departments
- Identify business goals and align UX designs
- Understand problem context and constraints

2. Conducting Product and In-depth User Research:

- Types: Qualitative (observable insights) and Quantitative (numerical data)
- Methods: Moderated (researcher present) vs Unmoderated
- Techniques: User interviews, usability testing, heatmaps, surveys
- Goal: Identify specific user issues and problem areas

3. Analyzing Research Results:

- Analyze information for key takeaways
- Identify pain points from qualitative research
- Understand metrics from quantitative research
- Develop user journey maps to visualize user flow
- Identify friction points to resolve

4. Designing Your Ideal Product:

- Create wireframes (skeletal framework)
- Develop prototypes (closer to final version)
- Test information architecture and navigation
- Apply UX design principles
- Enable testing of interaction and functionality

5. Testing and Noting Your Findings:

- Use different user testing methods
- Validate design through user feedback
- A/B testing for comparing alternatives
- Understand if developments meet user needs
- Document findings for iteration

6. Iterate and Continue Relevant Processes:

- UX research and design is never-ending cycle
- Products and users change over time
- Continuous improvement needed
- Every change should be made with users in mind
- Regular UX research to maintain quality

Q3.7 Describe the process of creating and using Empathy Maps with their merits and limitations.

Answer: Empathy Map Creation Process:

Definition: A collaborative visualization used to articulate what we know about a particular type of user, divided into four quadrants with the user in the middle.

Four Quadrants Explained:

1. **Says:** Direct quotes from user research, what users explicitly communicate
2. **Thinks:** Internal thoughts user might not vocalize, discovered through qualitative research
3. **Feels:** Emotional state, worries, and excitement during product experience
4. **Does:** Physical actions and behaviors observed during research

Creation Process:

1. Conduct user research (interviews, observations)
2. Collect data across all four quadrants
3. Organize findings into respective sections
4. Look for patterns and inconsistencies
5. Create individual maps first, then aggregate if needed

Merits:

- Helps understand user needs and goals accurately
- Shows complexity of user needs
- Easy to read and understand
- Great communication tool for design teams
- Ensures user-centered decision making
- Can collect data directly from users
- Cost-effective problem identification

Limitations:

- Not a replacement for journey mapping or user flows
- Cannot stand alone in design process
- May have ambiguous quadrant boundaries
- Possible inconsistencies between quadrants
- Requires skill to create meaningful insights

D. Application-Based Questions

Q3.8 Create an empathy map for designing a food delivery app for college students. Include at least 3 entries in each quadrant.

Answer: Empathy Map: College Students - Food Delivery App

User Profile: College student, age 19-22, budget-conscious, tech-savvy, busy schedule

SAYS:

- "I need food delivered fast between classes"
- "Can we split the order with roommates?"
- "Why is delivery fee so high for small orders?"
- "I want healthy options that are affordable"
- "The app should work even with slow wifi"

THINKS:

- "Should I spend money on delivery or cook?"
- "Hope the food arrives before my next class"
- "Wonder if this restaurant is actually good"
- "Maybe I can find a discount code"
- "I need to track my food expenses"

FEELS:

- **Hungry** - needs quick satisfaction
- **Anxious** - about spending limited budget
- **Excited** - when finding good deals
- **Frustrated** - with long delivery times
- **Overwhelmed** - by too many food options

DOES:

- Compares prices across different apps
- Reads reviews before ordering
- Shares orders with friends to reduce costs
- Uses university wifi to browse
- Orders late at night during study sessions
- Screenshots discount codes
- Tracks delivery on map constantly

Design Implications:

- Group ordering features
- Student discount integration
- Quick reorder options
- Budget tracking tools
- Offline menu browsing
- Clear delivery time estimates

Q3.9 You are tasked with improving the user experience of a hospital's patient registration system. Apply the 6-stage UX design process to this scenario.

Answer: Hospital Patient Registration System - 6-Stage UX Process:

Stage 1: Defining and Understanding Product

- **Stakeholder Interviews:** Hospital admin, reception staff, patients, doctors
- **Business Goals:** Reduce wait times, improve data accuracy, enhance patient satisfaction
- **Current Problems:** Long queues, repeated information entry, paper-based processes
- **Success Metrics:** Registration time, error rates, patient satisfaction scores

Stage 2: User Research

- **User Interviews:** Patients of different ages, emergency vs scheduled visits
- **Observational Study:** Current registration process flow
- **Staff Interviews:** Registration desk employees about pain points
- **Quantitative Data:** Average registration time, peak hours, error frequency

Stage 3: Analyzing Research Results

- **Key Pain Points:**
 - Lengthy forms with repetitive questions
 - Poor accessibility for elderly patients
 - No digital integration with medical records
 - Confusion about required documents
- **User Journey Map:** From hospital entry to successful registration
- **Personas:** Emergency patient, elderly patient, tech-savvy young adult

Stage 4: Design Solution

- **Wireframes:**
 - Self-service kiosk interface
 - Mobile pre-registration app
 - Simplified paper forms as backup
- **Information Architecture:** Logical flow from basic info to specific medical details
- **Prototypes:** Interactive kiosk simulation, mobile app mockup

Stage 5: Testing and Validation

- **Usability Testing:** Patients using prototype kiosk
- **A/B Testing:** Different form layouts
- **Accessibility Testing:** With visually impaired and elderly users
- **Staff Testing:** Registration desk workflow with new system

Stage 6: Implementation and Iteration

- **Phased Rollout:** Start with one department
- **Continuous Monitoring:** Registration times, user feedback
- **Regular Updates:** Based on seasonal patterns (flu season, etc.)
- **Staff Training:** Ongoing support for new system

Expected Outcomes:

- 50% reduction in registration time
- Improved data accuracy
- Better patient satisfaction
- Reduced staff workload

Q3.10 Design an empathy map for developing a fitness tracking app for working professionals aged 25-35. Consider their lifestyle and constraints.

Answer: Empathy Map: Working Professionals - Fitness Tracking App

User Profile: Working professional, 25-35 years, desk job, wants to stay fit but limited time

SAYS:

- "I don't have time to go to the gym"
- "I want to track my daily steps automatically"
- "Need quick workout routines for home"
- "Want to see my progress over time"
- "App should work without internet during commute"
- "I forget to log my workouts manually"

THINKS:

- "Am I getting enough exercise each day?"
- "Hope this app won't drain my phone battery"
- "Maybe I can exercise during lunch break"
- "Should set realistic goals I can actually achieve"
- "Wonder if my colleagues are more active than me"
- "Need to find motivation to stick to routine"

FEELS:

- **Guilty** - about sedentary lifestyle
- **Motivated** - when seeing progress charts
- **Frustrated** - when missing workout days
- **Competitive** - wants to compare with friends
- **Overwhelmed** - by complex fitness programs
- **Proud** - when achieving weekly goals
- **Tired** - after long work days

DOES:

- Checks phone frequently during work
- Takes stairs occasionally instead of elevator
- Parks farther away for extra steps
- Uses fitness apps during commute
- Works out early morning or late evening
- Shares achievements on social media
- Looks for quick 10-15 minute workouts
- Tracks steps with phone in pocket
- Sets multiple reminders to be active

Design Implications:

- Automatic activity tracking
- Quick home workout suggestions
- Battery-efficient design
- Social sharing features
- Flexible goal setting
- Offline workout content
- Integration with work calendar
- Reminder notifications
- Progress visualization
- Lunch break workout alerts

ADDITIONAL APPLICATION-BASED QUESTIONS

Healthcare Domain

Q4.1 Design a user interface for elderly patients to book doctor appointments online. Apply all relevant UX principles.

Answer: UX Principles Applied for Elderly Patient Appointment System:

1. Focus on the User:

- **User Research:** 65+ age group, limited tech experience, health concerns
- **Physical Considerations:** Potential vision/hearing impairments, slower motor skills
- **Mental Model:** Familiar with phone-based booking, prefer simple processes

2. Accessibility:

- **Large Text:** Minimum 18px font size, high contrast colors
- **Simple Navigation:** Clear, linear flow with minimal steps
- **Voice Support:** Text-to-speech for instructions
- **Multiple Input Methods:** Large buttons, voice input option

3. Consistency:

- **External:** Similar to familiar websites like banking portals
- **Internal:** Same button styles, colors, and layouts throughout

4. User Control:

- **Clear Back/Cancel:** Easy to return to previous step
- **Confirmation:** Show appointment details before final booking
- **Edit Options:** Ability to modify appointment details

5. Error Prevention:

- **Date Validation:** Prevent booking past dates or unavailable slots
- **Required Field Indicators:** Clear marking of mandatory information
- **Confirmation Dialogs:** "Are you sure?" for important actions

Design Features:

- Single-column layout
- Step-by-step progress indicator
- Large, clearly labeled buttons
- Emergency contact information visible
- Print option for appointment confirmation
- Phone number to call for help

E-commerce Domain

Q4.2 Evaluate the checkout process of an online shopping platform using heuristic principles. Provide detailed analysis.

Answer: Heuristic Evaluation: E-commerce Checkout Process

1. Visibility of System Status:

- **Current State:** Progress bar shows checkout steps
- **Issues:** Cart total not always visible, unclear shipping calculations
- **Recommendations:** Persistent cart summary, real-time price updates

2. Match Between System and Real World:

- **Good:** Shopping cart metaphor, checkout terminology
- **Issues:** Technical payment gateway terms
- **Recommendations:** Use familiar terms like "Continue Shopping" instead of "Return to Catalog"

3. User Control and Freedom:

- **Issues:** Difficult to edit items once in checkout, no save for later option
- **Recommendations:** Easy cart editing, guest checkout option, save cart feature

4. Consistency and Standards:

- **Issues:** Different button styles across steps, inconsistent form layouts
- **Recommendations:** Standardize button appearance, consistent error message formats

5. Error Prevention:

- **Good:** Address validation, card number formatting
- **Issues:** Allows invalid discount codes to be applied
- **Recommendations:** Real-time validation, suggested corrections

6. Recognition Rather than Recall:

- **Good:** Saved addresses, payment methods
- **Issues:** Must remember discount codes
- **Recommendations:** Show previously used discount codes, auto-apply eligible offers

7. Flexibility and Efficiency:

- **Good:** Express checkout for returning customers
- **Issues:** No keyboard shortcuts, limited customization
- **Recommendations:** One-click purchasing, customizable checkout fields

8. Aesthetic and Minimalist Design:

- **Issues:** Too much information on single page, promotional banners during checkout
- **Recommendations:** Focus on essential information only, remove distractions

9. Help Users Recognize and Recover from Errors:

- **Good:** Clear error messages for invalid cards
- **Issues:** Generic "Something went wrong" messages
- **Recommendations:** Specific error explanations, suggested fixes

10. Help and Documentation:

- **Issues:** No contextual help during checkout
- **Recommendations:** Inline help text, FAQ links, live chat support

Educational Technology

Q4.3 Create empathy maps for two different user types for an online learning platform: a) High school student, b) Working professional taking courses.

Answer: Empathy Map 1: High School Student - Online Learning Platform

SAYS:

- "I need help understanding this math concept"
- "Can I watch videos at 2x speed?"
- "When is this assignment due?"
- "My internet is slow at home"
- "I learn better with visual examples"

THINKS:

- "Hope this doesn't affect my grades"
- "Maybe I can study with friends online"
- "This is harder than classroom learning"
- "I need to manage my time better"
- "Wonder if teacher will notice if I'm multitasking"

FEELS:

- **Anxious** about online exams
- **Isolated** without classroom interaction
- **Distracted** by social media
- **Confused** about assignment requirements
- **Motivated** by gamified elements

DOES:

- Uses multiple devices simultaneously
 - Takes screenshots of important slides
 - Joins study groups on social platforms
 - Procrastinates on assignments
 - Asks questions in discussion forums
-

Empathy Map 2: Working Professional - Online Learning Platform

SAYS:

- "I need to upskill for career growth"
- "Can I download content for offline viewing?"
- "How long will this course take to complete?"
- "I need a certificate for my resume"
- "I only have evenings and weekends to study"

THINKS:

- "Will this course be worth the investment?"
- "How can I apply this to my current job?"
- "Need to balance work, family, and learning"
- "Hope the content is up-to-date with industry trends"
- "Should I take notes or just watch?"

FEELS:

- **Determined** to complete the course
- **Overwhelmed** by time constraints
- **Excited** about new knowledge
- **Pressure** to show immediate results
- **Frustrated** when technical issues occur

DOES:

- Studies during commute or breaks
- Takes detailed notes for later reference
- Searches for real-world applications
- Connects with other professionals in course
- Plans study schedule around work commitments

Design Implications:

- Mobile-responsive design for students
- Downloadable content for professionals
- Flexible pacing options
- Industry-relevant examples
- Certificate tracking
- Social learning features
- Multiple content formats (video, text, interactive)

This comprehensive question bank covers all the major concepts from your UX Design modules with a focus on application-based scenarios that test both theoretical knowledge and practical application skills.